

1)

a)

Given $\hat{X} = \frac{m}{N}$, $y_n \sim \text{Bernoulli}(x_0)$, $x_0 = P(Y=1)$

$$= \begin{cases} x_0 & y_n = 1 \\ 1-x_0 & y_n = 0 \end{cases}$$

$$m \sim \text{Binomial}(N, x_0) = \begin{cases} Nx_0 & y_n = 1 \\ 1-x_0 & y_n = 0 \end{cases}$$

Bias:

$$E[\hat{X}] = E\left[\frac{m}{N}\right] = \frac{1}{N} E[m] = \frac{1}{N} [Nx_0] = x_0$$

Since $E[\hat{X}] = x_0$, our estimator is unbiased

Variance:

$$\text{Var}(\hat{X}) = \frac{1}{N^2} \text{Var}(m) = \frac{1}{N^2} (Nx_0(1-x_0))$$

$\rightarrow \text{Var of Binomial dist}$

$$\boxed{\text{Var}(\hat{X}) = \frac{x_0(1-x_0)}{N}}$$

CRLB

$$\mathcal{L} = -m \ln(x_0) - (N-m) \ln(1-x_0)$$

$$\frac{\partial \mathcal{L}}{\partial x_0} = -\frac{m}{x_0} + \frac{N-m}{1-x_0} \quad \frac{\partial^2 \mathcal{L}}{\partial x_0^2} = \frac{m}{x_0^2} + \frac{N-m}{(1-x_0)^2}$$

$$J = E\left[\frac{m}{x_0^2} + \frac{N-m}{(1-x_0)^2}\right] = E\left[\frac{m}{x_0^2}\right] + E\left[\frac{N-m}{(1-x_0)^2}\right]$$

$$= \frac{1}{x_0^2} [Nx_0] + E\left[\frac{N}{(1-x_0)^2}\right] - E\left[\frac{m}{(1-x_0)^2}\right]$$

$$= \frac{N}{x_0} + \frac{N}{(1-x_0)^2} - \frac{Nx_0}{(1-x_0)^2} = \frac{N}{x_0} + \frac{N(1-x_0)}{(1-x_0)^2}$$

$$= \frac{N(1-x_0)^2}{x_0(1-x_0)^2} + \frac{Nx_0(1-x_0)}{x_0(1-x_0)^2}$$

$$= \frac{N[(1-x_0)^2 + x_0(1-x_0)]}{x_0(1-x_0)^2} = \frac{N[(1-x_0) + x_0]}{x_0(1-x_0)}$$

$$\bar{J} = \frac{N}{x_0(1-x_0)} \quad ; \quad \boxed{\bar{J}^{-1} = \frac{x_0(1-x_0)}{N}}$$

The estimator is consistent if

$$\lim_{N \rightarrow \infty} \text{Var}(\hat{x}) = 0$$

$$\lim_{N \rightarrow \infty} \frac{x_0(1-x_0)}{N} = 0, \text{ thus our estimator}$$

is consistent

Finally since the error variance achieves the CRLB,

$$\text{Var}(\hat{x}) = \frac{x_0(1-x_0)}{N} = \bar{J}^{-1}, \text{ the estimator is}$$

efficient

b)

$$\text{Given } \hat{x}_0 = \frac{\sum_{k=1}^N y_k V_k}{\sum_{k=1}^N V_k^2} \quad \text{where } V_k = e^{-a k \Delta t}$$

a const

Bias:

$$E[\hat{x}_0] = E\left[\frac{\sum_{k=1}^N y_k V_k}{\sum_{k=1}^N V_k^2} \right] = \frac{1}{\sum_{k=1}^N V_k^2} E\left[\sum_{k=1}^N y_k V_k \right]$$

$$y_k = x_k + V_k \quad x_k = x_0 V_k$$

$$y_k = V_k x_0 + V_k$$

$$\rightarrow \frac{1}{\sum_{k=1}^N V_k^2} E\left[\sum_{k=1}^N (V_k^2 x_0 + V_k^2) \right]$$

$$= \frac{E\left[\sum_{k=1}^N r_k^2 x_0\right] + E\left[\sum_{k=1}^N r_k\right]}{\sum_{k=1}^N r_k^2}$$

$$= \frac{\sum_{k=1}^N r_k^2 x_0 + 0}{\sum_{k=1}^N r_k^2} = \frac{x_0 \sum_{k=1}^N r_k^2}{\sum_{k=1}^N r_k^2} = \boxed{x_0}$$

Since $E[\hat{x}_0] = x_0$, our estimate is unbiased

$$\text{Var}(\hat{x}_0) = \text{Var}\left(\frac{\sum_{k=1}^N r_k^2 x_0 + r_k v_k}{\sum_{k=1}^N r_k^2}\right)$$

$$= \frac{\text{Var}\left(\sum_{k=1}^N r_k^2 x_0 + r_k v_k\right)}{\left[\sum_{k=1}^N r_k^2\right]^2} = \frac{\text{Var}\left(\sum_{k=1}^N r_k^2 x_0 + \sum_{k=1}^N r_k v_k\right)}{\left[\sum_{k=1}^N r_k^2\right]^2}$$

$$= \frac{\text{Var}\left(\sum_{k=1}^N r_k v_k\right)}{\left[\sum_{k=1}^N r_k^2\right]^2}$$

Since v_k is AWGN, v_k are independent

$$= \frac{\sum_{k=1}^N r_k^2 \text{Var}(v_k)}{\left[\sum_{k=1}^N r_k^2\right]^2} = \boxed{\frac{\sigma^2}{\sum_{k=1}^N r_k^2}}$$

CRLB:

$$p(y_0, y) = \frac{1}{\sqrt{M}} \exp\left[-\frac{1}{2} \sum_{k=1}^N |y_k - r_k x|^2\right]$$

$$\Lambda(x_0; y_{1:N}) = \frac{1}{(2\pi)^{N/2} \sigma^N} \exp \left[-\frac{1}{2\sigma^2} \sum_{k=1}^N (y_k - r_k x_0)^2 \right]$$

$$\ln \Lambda = \ln \left(\frac{1}{(2\pi)^{N/2} \sigma^N} \right) - \frac{1}{2\sigma^2} \sum_{k=1}^N (y_k - r_k x_0)^2$$

$$y_k = r_k x_0 + v_k$$

$$\frac{\partial \ln \Lambda}{\partial x} = \frac{1}{2\sigma^2} \sum_{k=1}^N 2(y_k - r_k x_0) r_k$$

$$= \frac{1}{\sigma^2} \sum_{k=1}^N r_k y_k$$

$$J = E \left[\left(\frac{1}{\sigma^2} \sum_{k=1}^N r_k y_k \right)^2 \right]$$

$$= \frac{1}{\sigma^4} E \left[\sum_{k=1}^N r_k^2 v_k^2 \right] = \frac{1}{\sigma^4} \sum_{k=1}^N r_k^2 E[v_k^2]$$

$$= \frac{\left[\sum_{k=1}^N r_k^2 \right] \text{Var}(v_k)}{\sigma^4} = \frac{\sum_{k=1}^N r_k^2}{\sigma^2}$$

Thus: $J^{-1} = \frac{\sigma^2}{\sum_{k=1}^N r_k^2} = \text{Var}(\hat{x}_0)$

Since $\text{Var}(\hat{x}_0) = J^{-1} = \frac{\sigma^2}{\sum_{k=1}^N r_k^2}$, the estimator is

efficient

c)

$$y_k = \mu + v_k \quad v_k \sim (0, \sigma^2)$$

$$\Lambda(X; Y_{1:N}) = P(Y_{1:N} | X), \quad X = \begin{bmatrix} \mu \\ \sigma^2 \end{bmatrix}$$

Given Y_k i.i.d., $P(Y_{1:N} | X) = \prod_{k=1}^N P(Y_k | X)$

$$P(Y_k) = N(0, \sigma^2) \propto \exp\left(-\frac{1}{2\sigma^2} Y_k^2\right)$$

$$Y_k = \mu + V_k \quad V_k = Y_k - \mu$$

$$P(Y_k - \mu) \propto \exp\left(-\frac{1}{2\sigma^2} (Y_k - \mu)^2\right) = P(Y_k | X)$$

$$\Lambda(X; Y_{1:N}) = \prod_{k=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} [Y_k - \mu]^2\right)$$

$$= \frac{1}{(2\pi)^{N/2} \sigma^N} \exp\left[-\frac{1}{2\sigma^2} \sum_{k=1}^N (Y_k - \mu)^2\right]$$

$$\mathcal{L}(X; Y_{1:N}) = -\ln \Lambda(X; Y_{1:N})$$

$$= -\ln\left(\frac{1}{(2\pi)^{N/2} \sigma^N}\right) + \frac{1}{2\sigma^2} \sum_{k=1}^N (Y_k - \mu)^2$$

$$\nabla_{X_0} \mathcal{L}(X; Y_{1:N}) = \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial \mu} \\ \frac{\partial \mathcal{L}}{\partial \sigma^2} \end{bmatrix}_{\hat{X}}$$

$$\frac{\partial \mathcal{L}}{\partial \mu} \Big|_{\hat{\mu}} = -\frac{1}{\sigma^2} \sum_{k=1}^N Y_k - \hat{\mu} = 0$$

$$\rightarrow \sum_{k=1}^N Y_k - \sum_{k=1}^N \hat{\mu} = 0 \rightarrow \hat{\mu} = \frac{\sum_{k=1}^N Y_k}{N}$$

$$\frac{\partial \mathcal{L}}{\partial \sigma^2} \Big|_{\hat{\sigma}^2} = \frac{1}{(2\pi)^{N/2} \hat{\sigma}^N} \left(\frac{N}{2} (2\pi)^{N/2} \hat{\sigma}^{N-2} \right) - \frac{1}{2\hat{\sigma}^4} \sum_{k=1}^N (Y_k - \hat{\mu})^2 = 0$$

$$\sum_{k=1}^N (Y_k - \hat{\mu})^2 = \frac{(N/2) (2\pi)^{N/2} \hat{\sigma}^{N-2}}{(2\pi)^{N/2} \hat{\sigma}^N} \cdot 2\hat{\sigma}^4$$

$$= \frac{N \hat{\sigma}^N \hat{\sigma}^2}{\hat{\sigma}^N}$$

$$= \frac{N \hat{\sigma}^2 \hat{\sigma}^2}{\hat{\sigma}^2}$$

$$\sum_{k=1}^N (y_k - \hat{\mu})^2 = N \hat{\sigma}^2$$

$$\hat{\sigma}^2 = \frac{\sum_{k=1}^N (y_k - \hat{\mu})^2}{N}$$

Unknown μ , known σ^2

$$E[\hat{\mu}] = E\left[\frac{1}{N} \sum_{k=1}^N y_k\right] = \frac{1}{N} E\left[\sum_{k=1}^N \mu + v_k\right]$$

$$= \frac{1}{N} \left[E[N\mu] + \underbrace{E\left[\sum_{k=1}^N v_k\right]}_0 \right] = \boxed{\mu} \quad (\text{ unbiased })$$

$$\text{Var}(\hat{\mu}) = E[(\hat{\mu} - \mu)^2] = E\left[\left(\frac{1}{N} \sum_{k=1}^N y_k - \mu\right)^2\right]$$

$$= E\left[\left(\frac{1}{N} \sum_{k=1}^N y_k\right)^2\right] - \mu^2 = E\left[\left(\frac{1}{N} \sum_{k=1}^N (\mu + v_k)\right)^2\right] - \mu^2$$

$$= E\left[\left(\frac{1}{N} \sum_{k=1}^N \mu + \frac{1}{N} \sum_{k=1}^N v_k\right)^2\right] - \mu^2$$

$$= E\left[\left(\mu + \frac{1}{N} \sum_{k=1}^N v_k\right)^2\right] - \mu^2 = E\left[\mu^2 + \frac{2\mu}{N} \sum_{k=1}^N v_k + \frac{1}{N^2} \sum_{k=1}^N v_k^2\right] - \mu^2$$

$$= \mu^2 + \frac{2\mu}{N} E\left[\sum_{k=1}^N v_k\right] + \frac{1}{N^2} E\left[\sum_{k=1}^N v_k^2\right] - \mu^2$$

$$= \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}$$

Unknown σ^2 , known μ

$$E[\hat{\sigma}^2] = E\left[\frac{1}{N} \sum_{k=1}^N [y_k - \mu]^2\right] = E\left[\frac{1}{N} \sum_{k=1}^N [\mu v_k - \mu]^2\right]$$

$$= E\left[\frac{1}{N} \sum_{k=1}^N v_k^2\right] = \boxed{\sigma^2}$$

$$\text{Var}(\hat{\sigma}^2) = \text{Var}\left(\frac{1}{N} \sum_{k=1}^N v_k^2\right) = \frac{1}{N^2} \sum_{k=1}^N \text{Var}(v_k^2)$$

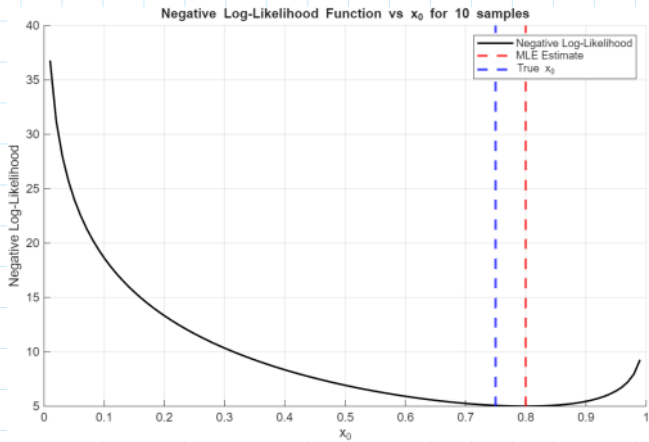
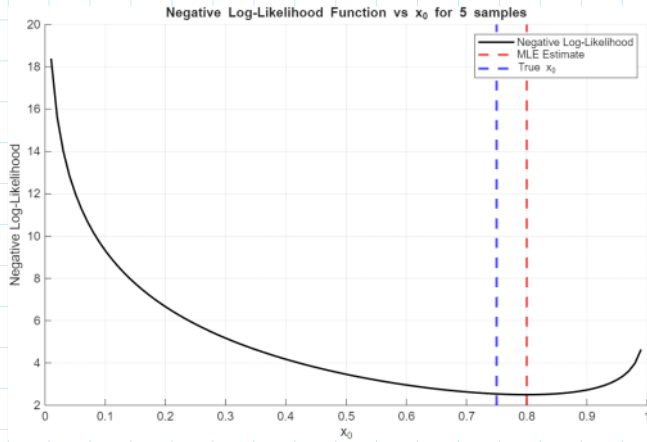
$$= \frac{1}{N} \sum_{n=1}^N E[V_n^4] - (E[V_n^2])^2 \quad E[V_n^2] = \text{Var}(V_n) = \sigma^2$$

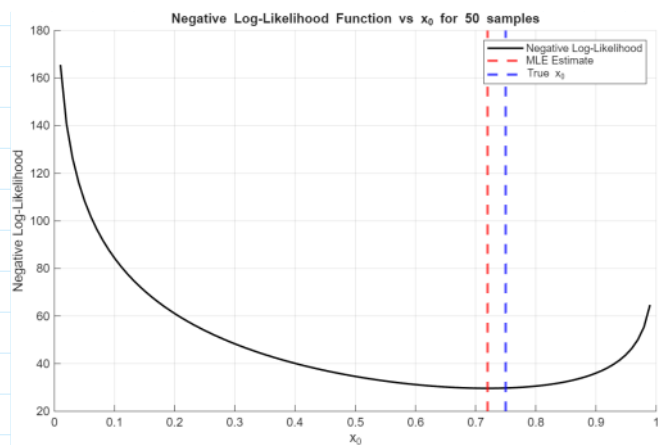
$$E[V_n^4] = 3\sigma^4$$

$$= \frac{1}{N} [3N\sigma^4 - N\sigma^4] = \boxed{\frac{2\sigma^4}{N}}$$

2) Code at the bottom of page.

a)





It is clear that the MLE estimate corresponds with the minimum value of L .

b)

$$\text{Mean}(\hat{x}_0^j) = 0.7503$$

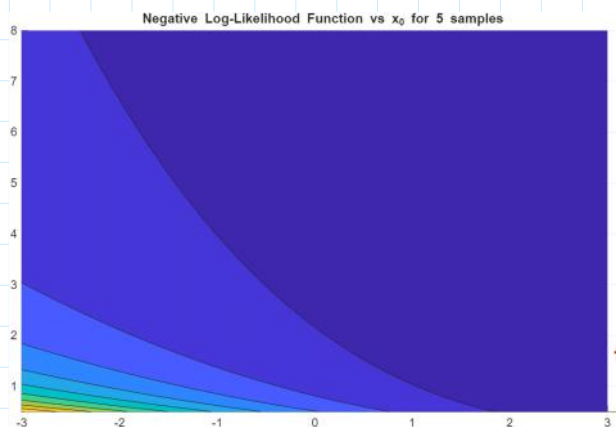
$$\text{Var}(\hat{x}_0^j) = 0.0189$$

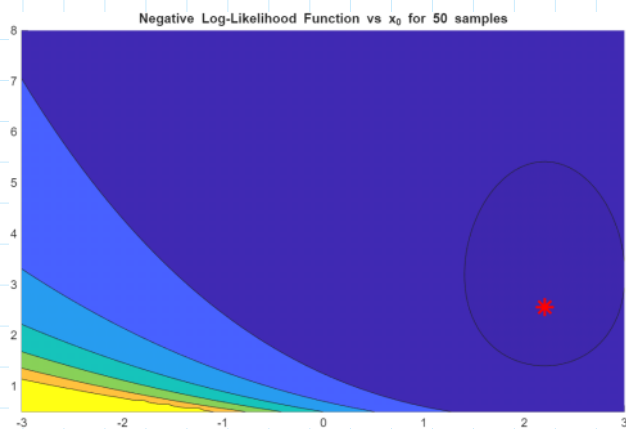
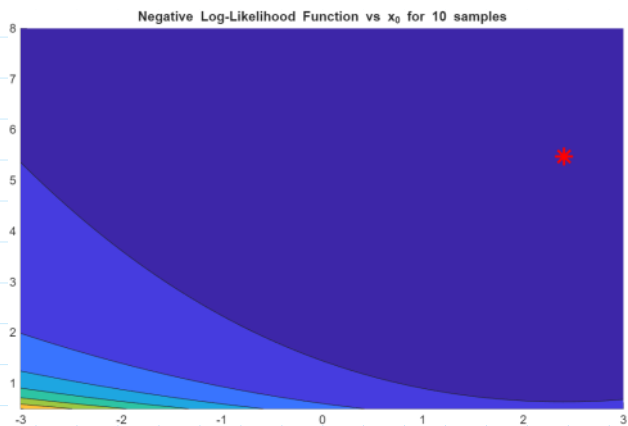
We see the $\text{Mean}(\hat{x}_0^j) \rightarrow x_0 = 0.75$, which agrees with the computed bias of 0.

$$\text{For } N=10, \text{CRLB} = \frac{x_0(1-x_0)}{10} = 0.01875$$

$\text{Var}(\hat{x}_0^j) > \text{CRLB}$ which is what we expect for an unbiased estimator

c)





It is clear from the plots that the minimum of J corresponds with the MLE estimate.

d) Unknown μ , known σ

$$\text{Mean}(\hat{\mu}_j) = 2.0088$$

$$\text{Var}(\hat{\mu}_j) = 0.3033$$

$\text{Mean}(\hat{\mu}_j) \rightarrow \mu = 2$, which agrees with the expected bias of zero.

CRLB:

$$J = -E \left[\frac{\partial^2}{\partial \mu^2} \ln \Lambda(x_0; y_{1:N}) \right]$$

$$-\ln \Lambda(x_0; y_{1:N}) = -\ln \left(\frac{1}{(2\pi)^{N/2} \sigma^N} \right) + \frac{1}{2\sigma^2} \sum_{n=1}^N (y_n - \mu)^2 = J$$

$$\frac{\partial J}{\partial \mu} = -\frac{1}{\sigma^2} \sum_{n=1}^N (y_n - \mu) \quad \frac{\partial^2 J}{\partial \mu^2} = \frac{1}{\sigma^2} \sum_{n=1}^N -1 = -\frac{N}{\sigma^2}$$

$$J = -E \left[-\frac{N}{\sigma^2} \right] = \frac{N}{\sigma^2}, \text{ for } \sigma^2 \text{ assumed known}$$

$$J^{-1} = \frac{\sigma^2}{N} \rightarrow \text{CRLB} \quad J^{-1} = \text{Var}(\hat{\mu})$$

Expect to see $\text{Var}(\hat{\mu}_j) \geq J^{-1}$

$$J^{-1} = \frac{3}{10} = 0.3 \quad 0.3053 \geq 0.3$$

This our variance agrees w/ the empirical CRLB.



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Bijan Jourabchi

ASEN 6044 HW1 Q2

```
clear; clc; close all
rng(100)
```

Part a)

```
x0 = 0.75;
N = 5;
bernoulliParameterEstimationSim(x0, N, true);

N = 10;
bernoulliParameterEstimationSim(x0, N, true);

N = 50;
bernoulliParameterEstimationSim(x0, N, true);
```

Part b)

```
x_hat_vec = zeros(1,5000);
for j = 1:5000
    x_hat_vec(j) = bernoulliParameterEstimationSim(x0, 10, false);
end
avg = mean(x_hat_vec)
variance = var(x_hat_vec)
```

Part c)

```
x0 = [2,3];
unknownScalarMeans(x0, 5, true);
```

```
unknownScalarMeas(x0, 10, true);
unknownScalarMeas(x0, 50, true);
```

Part d)

```
x_hat_vec = zeros(2,5000);
for j = 1:5000
    x_hat_vec(:,j) = unknownScalarMeas(x0, 10, false);
end

avg = mean(x_hat_vec(1,:))
variance = var(x_hat_vec(1,:))

function [x_hat] = bernoulliParameterEstimationSim(x0, N, PLOT)

sample = rand(N,1);
Y = sample <= x0;

% results
m = sum(Y);

% MLE Result
x_hat = m/N;
% Likelihood curve

x = linspace(0,1,100);
L = -n*log(x) - (N-m)*log(1-x);

if PLOT
    fprintf("x_MLE = %.2f for %d samples \n",x_hat,N)
    figure(N); hold on; grid on

    plot(x,L,'b',LineWidth=1.5)
    xline(x_hat,'r--',LineWidth=2)
    xline(x0,'b--',LineWidth=2)

    title("Negative Log-Likelihood Function vs x_0 for " + N + " samples")
    xlabel("x_0")
    ylabel("Negative Log-Likelihood")

    legend(["Negative Log-Likelihood", "MLE Estimate", "True x_0"])
end
end

function [x_hat] = unknownScalarMeas(x0, N, PLOT)

% unpack
mu = x0(1);
sigmaqr = x0(2);
```

2

```
% Sample
Y = normrnd(mu,sqrt(sigmaqr),1,N);

% MLN results
mu_hat = sum(Y)/N;
temp = (Y - mu_hat).^2;
sigmaqr_hat = sum(temp) / N;

x_hat = [mu_hat; sigmaqr_hat];

% Negative Log Likelihood
sigmaqrRange = linspace(0.5,8,100);
muRange = linspace(-3,3,100);
L = zeros(100,100);

for i = 1:100
    for j = 1:100
        L(i,j) = -log(1/(2*pi*sigmaqrRange(j))^(N/2)) * exp((-1/(2*sigmaqrRange(j)) * sum((Y-muRange(i)).^2)));
    end
end

if PLOT
    figure(N+1); hold on; grid on
    contour(muRange, sigmaqrRange, L)
    plot(mu_hat, sigmaqr_hat, 'r', 'MarkerSize', 15, 'LineWidth', 2)
    title("Negative Log-Likelihood Function vs x_0 for " + N + " samples")
end
end

x_MLE = 0.80 for 5 samples
x_MLE = 0.80 for 10 samples
x_MLE = 0.72 for 50 samples

avg =

    0.7503

variance =

    0.0189

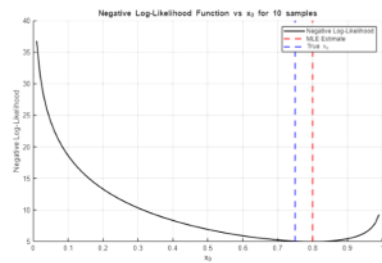
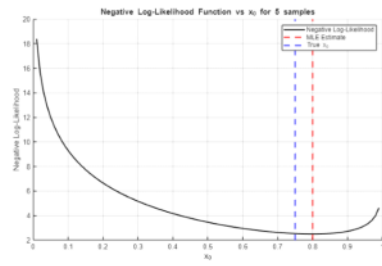
avg =

    2.0088

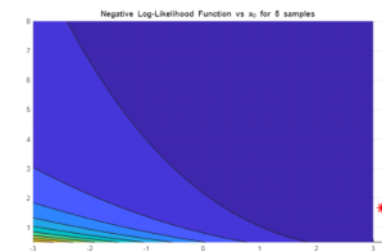
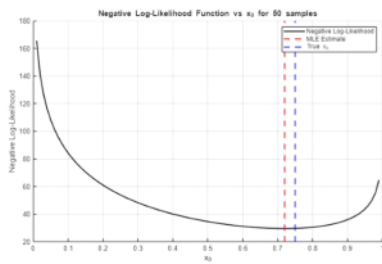
variance =
```

3

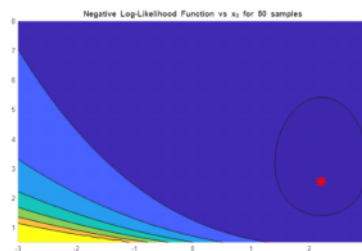
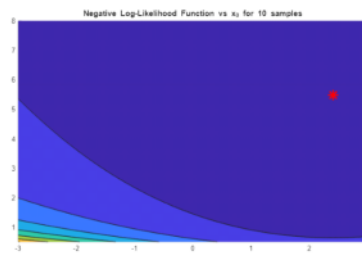
0.3033



4



5



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3)

$$a) P(y_n | \theta) = \begin{cases} \frac{y^n e^{-\frac{y}{\theta}}}{n! (\theta)^{n+1}} & y \geq 0 \\ 0 & y < 0 \end{cases}$$

y i.i.d:

$$y^n e^{-\frac{y}{\theta}} \quad y \geq 0$$

y i.i.d:

$$P(Y_{1:N}|\theta) = \prod_{k=1}^N \begin{cases} \frac{y_k^m e^{-\frac{y_k}{2\theta}}}{m! (2\theta)^{m+1}} & y_k \geq 0 \\ 0 & y_k < 0 \end{cases} = \Lambda(\theta; Y_{1:N})$$

$$\mathcal{L}(Y_{1:N}; \theta) = -\log(\Lambda(\theta; Y_{1:N}))$$

for $y \geq 0$

$$\mathcal{L} = -\log\left(\prod_{k=1}^N \frac{y_k^m e^{-\frac{y_k}{2\theta}}}{m! (2\theta)^{m+1}}\right)$$

$$= -\sum_{k=1}^N \log\left(\frac{y_k^m e^{-\frac{y_k}{2\theta}}}{m! (2\theta)^{m+1}}\right)$$

$$= -\sum_{k=1}^N \left[\log(y_k^m e^{-\frac{y_k}{2\theta}}) - \log(m! (2\theta)^{m+1}) \right]$$

$$= N \log(m! (2\theta)^{m+1}) - \sum_{k=1}^N \left[\log(y_k^m e^{-\frac{y_k}{2\theta}}) \right]$$

$$= N \log(m! (2\theta)^{m+1}) - \left[\sum_{k=1}^N \log(y_k^m) + \sum_{k=1}^N \left(-\frac{y_k}{2\theta}\right) \right]$$

$$\mathcal{L} = N \log(m!) + N \log[(2\theta)^{m+1}] - \sum_{k=1}^N \log(y_k^m) + \sum_{k=1}^N \frac{y_k}{2\theta}$$

$$\frac{d\mathcal{L}}{d\theta} = \frac{2N}{(2\theta)^{m+1}} (m+1) (2\theta)^m - \frac{1}{2\theta^2} \sum_{k=1}^N y_k = 0$$

$$\frac{2N}{2\theta} (m+1) = \frac{1}{2\theta^2} \sum_{k=1}^N y_k$$

$$\frac{2\theta^2}{\theta} = \frac{1}{N(m+1)} \sum_{k=1}^N y_k$$

$$\hat{\theta}_{ML} = \frac{1}{2N(m+1)} \sum_{k=1}^N y_k$$

b)

$$E[\hat{\theta}_{ML} - \theta] = E[\hat{\theta}] - \theta$$

$$E[\hat{\theta}] = \frac{1}{2N(m+1)} E\left[\sum_{k=1}^N y_k\right] = \frac{1}{2N(m+1)} \sum_{k=1}^N E[y_k]$$

$$= \frac{E[y_k]}{2(m+1)} \quad \text{Recognize } y_k \sim \text{Gamma}(m+1, 2\theta)$$

$$\text{Thus, } E[y_k] = 2\theta(m+1)$$

$$E[\hat{\theta}] = \theta, \quad \text{so, } E[\hat{\theta} - \theta] = 0, \quad \text{estimator is unbiased.}$$

$$\text{Var}(\hat{\theta}) = \text{Var}\left(\frac{1}{2N(m+1)} \sum_{k=1}^N y_k\right) = \frac{1}{[2N(m+1)]^2} \text{Var}\left(\sum_{k=1}^N y_k\right)$$

$$\text{Since } y_k \text{ i.i.d., } \text{Var}\left(\sum_{k=1}^N y_k\right) = N \text{Var}(y_k) = N(m+1)(2\theta)^2$$

$$\text{Var}(\hat{\theta}) = \frac{N(m+1)4\theta^2}{4N^2(m+1)^2} = \boxed{\frac{\theta^2}{N(m+1)}}$$

c)

$$J = E\left[\left(\frac{\partial}{\partial \theta} \ln \Lambda(x_0; y_{1:N})\right) \left(\frac{\partial}{\partial \theta} \ln \Lambda(x_0; y_{1:N})\right)^T\right]$$

$$\ln \Lambda = -N \log(m!) - N \log[(2\theta)^{m+1}] + \sum_{k=1}^N \log(y_k) - \sum_{k=1}^N \frac{y_k}{2\theta}$$

$$\frac{\partial}{\partial \theta} \ln \Lambda = \frac{-2N(m+1)(2\theta)^m}{(2\theta)^{m+1}} + \sum_{k=1}^N \frac{y_k}{2\theta^2}$$

$$= \frac{-N(m+1)}{\theta} + \sum_{k=1}^N \frac{y_k}{2\theta^2}$$

$$J = E\left[\left(\frac{\partial}{\partial \theta} \ln \Lambda\right)^2\right] = -E\left[\frac{\partial^2}{\partial \theta^2} \ln \Lambda\right]$$

$$\frac{\partial^2}{\partial \theta^2} \ln \Lambda(x_0; y_{1:N}) = \frac{N(m+1)}{\theta^2} - \sum_{k=1}^N \frac{y_k}{\theta^3}$$

$$-E\left[\frac{N(m+1)}{\theta^2} - \sum_{k=1}^N \frac{y_k}{\theta^3}\right] = \frac{-N(m+1)}{\theta^2} + \frac{N}{\theta^3} E[y_k]$$

$$-E \left[\frac{N(m+1)}{\theta^2} - \sum_{k=1}^N \frac{y_k}{\theta^3} \right] = \frac{-N(m+1)}{\theta^2} + \frac{N}{\theta^3} E[y_k]$$

$$= \frac{-N(m+1)}{\theta^2} + \frac{N(m+1)(2\theta)}{\theta^3} = \frac{-N(m+1)}{\theta^2} + \frac{2N(m+1)}{\theta^2}$$

$$J = \frac{N(m+1)}{\theta^2} \rightarrow \text{CRLB} = J^{-1} = \boxed{\frac{\theta^2}{N(m+1)}}$$

d) Since our estimator is unbiased, it follows that it is efficient since

$$\text{Var}(\hat{\theta}) = \text{CRLB} = \frac{\theta^2}{N(m+1)}$$

$\lim_{N \rightarrow \infty} \frac{\theta^2}{N(m+1)} = 0$, thus our estimator is consistent.

4) Code at the end of problem

a)

For the measurement model:

$$h(x_0) = \begin{bmatrix} \sqrt{(L - g(t))^2 + a(t)^2} + v_r(t) \\ \tan^{-1}\left(\frac{a(t)}{L - g(t)}\right) + v_b(t) \end{bmatrix}$$

We need to take a derivative of $h(x_0)$ w.r.t x_0 in order to implement the Gauss-Newton algorithm for NLS.

Since $g(t)$ & $a(t)$ are functions of x_0 , we can rewrite $h(x_0)$ as such:

$$h(x_0) = \begin{bmatrix} \sqrt{[L - (g(0) + \dot{g}(0)t)]^2 + [a(0) + \dot{a}(0)t - \frac{1}{2}gt^2]^2} + v_r(t) \\ \tan^{-1}\left(\frac{a(0) + \dot{a}(0)t - \frac{1}{2}gt^2}{L - g(0) - \dot{g}(0)t}\right) + v_b(t) \end{bmatrix} = \begin{bmatrix} p(x_0) \\ \phi(x_0) \end{bmatrix}$$

$\frac{\partial h(x_0)}{\partial x_0}$ will be 2×4 Jacobian matrix.

$$\frac{\partial h(x_0)}{\partial x_0} = \begin{bmatrix} \frac{\partial p}{\partial s_0} & \frac{\partial p}{\partial \dot{s}_0} & \frac{\partial p}{\partial a_0} & \frac{\partial p}{\partial \dot{a}_0} \\ \frac{\partial \theta}{\partial s_0} & \frac{\partial \theta}{\partial \dot{s}_0} & \frac{\partial \theta}{\partial a_0} & \frac{\partial \theta}{\partial \dot{a}_0} \end{bmatrix}$$

By symbolically taking the partial derivatives, we get

$$\frac{\partial h(x_0)}{\partial x_0} = \begin{bmatrix} \frac{2(\dot{s}_0 t + s_0 - L)}{\sqrt{(2a_0 - gt^2 + 2\dot{a}_0 t)^2 + 4(\dot{s}_0 t + s_0 - L)^2}} & \frac{2t(\dot{s}_0 t + s_0 - L)}{\sqrt{(\dots)}} & \frac{2a_0 - gt^2 + 2\dot{a}_0 t}{\sqrt{\dots}} & \frac{t(2a_0 - gt^2 + 2\dot{a}_0 t)}{\sqrt{\dots}} \\ \frac{2(2a_0 - gt^2 + 2\dot{a}_0 t)}{(2a_0 - gt^2 + 2\dot{a}_0 t)^2 + 4(\dot{s}_0 t + s_0 - L)^2} & \frac{2t(2a_0 - gt^2 + 2\dot{a}_0 t)}{(\dots)} & \frac{4(-\dot{s}_0 t - s_0 + L)}{\dots} & \frac{-4t(-\dot{s}_0 t - s_0 + L)}{\dots} \end{bmatrix}$$

This can be evaluated at each time step t_k w/ $\hat{x}_{0,NLS}$ to implement the Gauss-Newton algorithm.

b) To adjust the α step size, I used the same method presented in class:

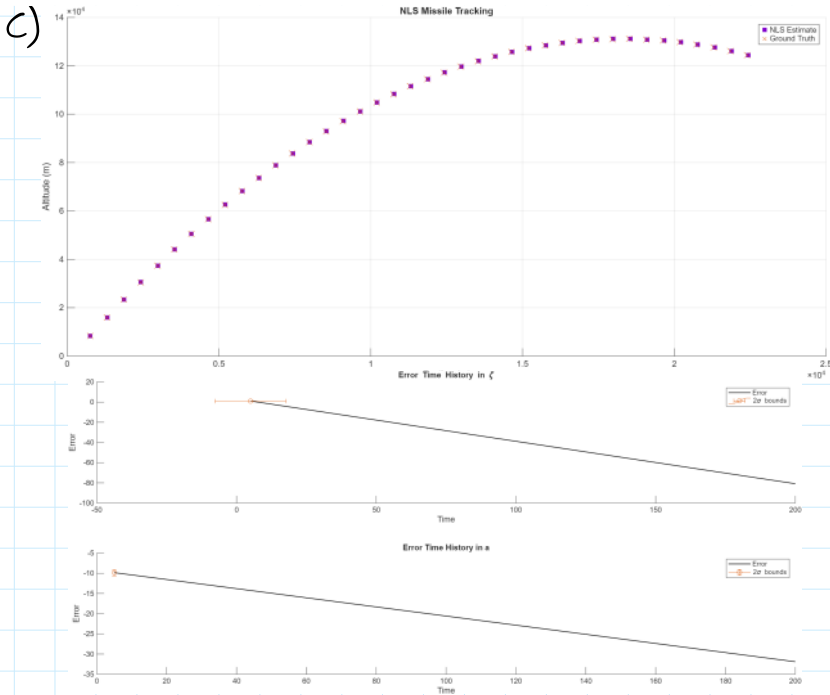
If: $J_n > J_c$: $\alpha = \alpha/2$

Else: $\hat{x}_c = \hat{x}_n$

To check for convergence, I checked when the difference between the previous iteration & current iteration's cost function changed by a value less than ϵ so:

if $|J_{prev} - J_{curr}| < \epsilon$: Converged = true





Converged in 52 iterations.

$$\hat{x}_{NLS} = \begin{bmatrix} 207.91 \\ 111.1808 \\ 474.1528 \\ 1.604 \times 10^3 \end{bmatrix}$$

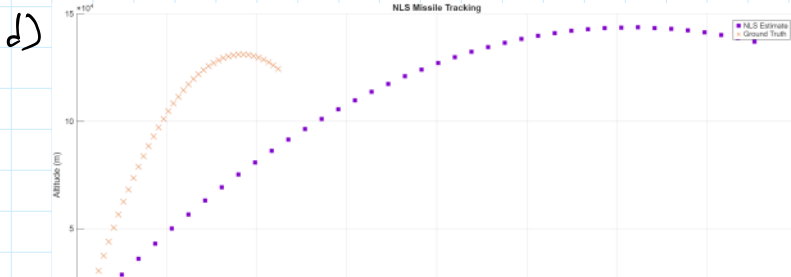
The associated error covariance is given by

$$P(\hat{x}_{NLS}) \approx H^T(\hat{x}_{NLS}) V(\hat{x}_{NLS})$$

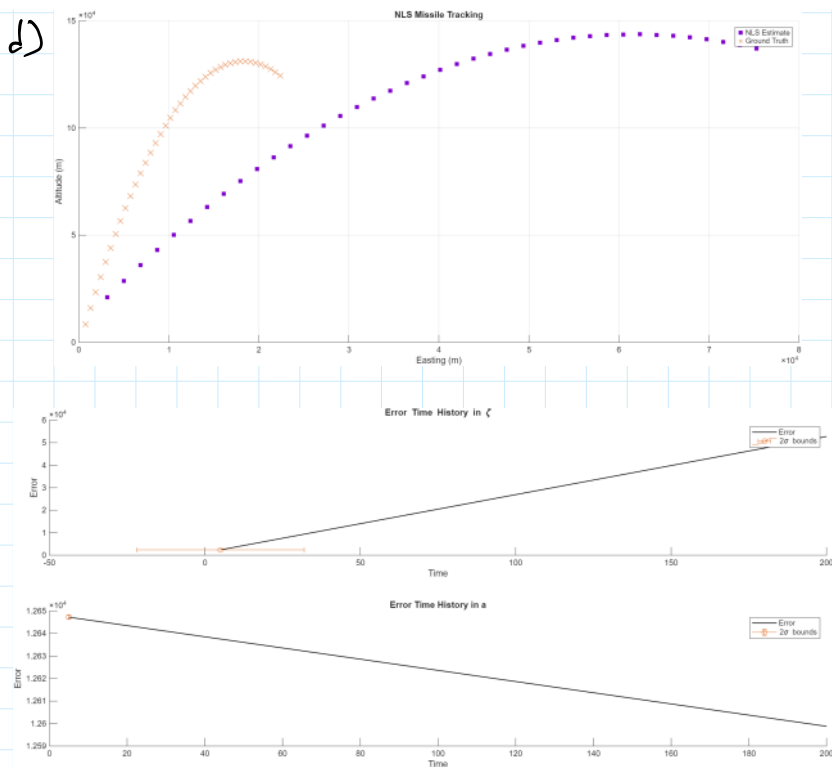
for "small errors" residuals and was computed to be:

$$P(\hat{x}_{NLS}) = \begin{bmatrix} 40.5 & -2.76 & -55.11 & -0.81 \\ -2.76 & 0.61 & 20.81 & 0.16 \\ -55.11 & 20.81 & 185.4 & 4.98 \\ -0.81 & 0.16 & 4.98 & 0.043 \end{bmatrix}$$

Based off the plot and assumptions of the problem our estimate for x_0 seems accurate, and indeed it is, with $\|\hat{x}_0 - x_{true}\| = 19.66$ showing the fault is estimate is not far off.



d)



Converged in 16 iterations, $\hat{X}_{NLS} \approx \begin{bmatrix} 1.31 \times 10^3 \\ 389.97 \\ 1.31 \times 10^4 \\ 1.6 \times 10^3 \end{bmatrix}$

$$P(\hat{X}_{NLS}): \begin{bmatrix} 182.15 & 9.12 & 628.19 & -2.62 \\ 9.12 & 0.6047 & 38.34 & -0.155 \\ 628.19 & 38.44 & 2.5 \times 10^3 & -10.28 \\ -2.62 & -0.155 & -10.15 & 0.0424 \end{bmatrix}$$

For this initial guess of $\hat{X}_0$, our algorithm does not converge to the correct X_0 . Because of this, our approximation of $P(\hat{X}_{NLS})$ is likely far off the true covariance, as the measurement residuals are not small, and thus $P(\hat{X}_{NLS}) \approx H^T H$ is not a good estimate.

Finally, $\|\hat{X}_0 - X_c\| = 1.2138 \times 10^4$, further showing how inaccurate $\hat{X}_0$ is as an estimate.

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Bijan Jourabchi

ASEN 6044 HW1 Q4

Load Datafile

```
clear; clc; close all
load('homework1_missileprob_data.mat')
```

Get Noise Covar matrix

```
N = size(ystacked,1)/2;
np = 2;
L = 80000;
dt = 0.5; % sec
t = dt*dt:N*dt;
```

```
R = [70 0; 0 0.005];
Rb = kron(eye(N), R);
```

Whitening transformation

```
S = chol(Rb);
ya = S*ystackeds;
```

NLS

```
x0g = [10;40;2300;30];
%x0g = [130;4;2;6050];

% NLS
itr = 0;
maxitr = 100;
```

1

```
maxalph = 10;
converged = false;

while itr < maxitr && ~converged
    % Setup NLS
    yc = zeros(np,N);
    R = zeros(np*N,4);

    for k = 1:N
        yc(:,k) = h(x0g,t(k),R, L);

        % [xi,a] = f(x0g,t(k));
        H(2*k-1 : 2*k, :) = jcb(x0g,L,t(k));

    end
    yc = S*yc();
    R = S*R;

    % Cost fn
    Jcurr = (ya - yc)'/Rb*(ya - yc);
    dx = inv(R'*inv(Rb)*R)*R'*inv(Rb)*(ya - yc);
    alph = 1;
    alphapits = 0;
    while alphapits < maxalph
        xn0 = x0g + alph*dx;

        yc = zeros(np,N);
        for k = 1:N
            yc(:,k) = h(xn0,t(k),R, L);
        end
        yc = S*yc();

        Jn = (ya - yc)'/Rb*(ya - yc);

        if abs(Jcurr - Jn) < 50
            converged = true;
            break;
        end

        if Jn > Jcurr
            alph = alph/2;
        else
            x0g = xn0;
            break;
        end
        alphapits = alphapits + 1;
    end
    itr = itr + 1;
end

plot_NLS(x0true,x0g,ystackeds,t,Rb)
```

2

Measurment function

```
function [y] = h(x0,t,R, L)

    vk = mvnrand([0,0],R)';

    [xik,ak] = f(x0,t);

    rho = sqrt((L - xik)^2 + ak^2) + vk(1);
    theta = atan2(ak, (L-xik)) + vk(2);

    y = [rho;theta];
end
```

Dynamics function

```
function [xik,ak] = f(x0,t)

    x10 = x0(1);
    x1d0 = x0(2);
    a0 = x0(3);
    ad0 = x0(4);
    g = 9.81;

    xik = x10 + x1d0*t;
    ak = a0 + ad0*t - 0.5*g*t^2;
end
```

Jacobian fxn

```
function [H] = jcb(x,L,t)

    x1 = x(1);
    x1d = x(2);
    a = x(3);
    ad = x(4);
    g = 9.81;

    H11 = 2*(xid*t*x1-L)/sqrt((2*a-g*t^2+2*ad*t)^2 + 4*(xid*t+x1-L)^2);
    H12 = H11 * t;
    H13 = (2*a - g*t^2 + 2*ad*t) / sqrt((2*a-g*t^2+2*ad*t)^2 + 4*(xid*t+x1-L)^2);
    H14 = H13*t;

    H21 = 2*(2*a - g*t^2 + 2*ad*t)/((2*a - g*t^2 +2*ad*t)^2 + 4*(xid*t+x1-L)^2);
    H22 = H21*t;
    H23 = 4*(-xid*t-x1+L)/((2*a - g*t^2 +2*ad*t)^2 + 4*(xid*t+x1-L)^2);
    H24 = -t*H23;
```

3

```
H = [H11 H12 H13 H14;H21 H22 H23 H24];
```

```
end
```

Plotting Functions

```
function [] = plot_MLS(x0t,x0MLS,Y,tvec,R)

    N = length(tvec);

    % Get Truth/Estimated Traj
    trajT = zeros(2,N);
    trajE = zeros(2,N);

    for k = 1:N
        [x1,a] = f(x0t,tvec(k));
        trajT(1,k) = [x1;a];
        [x1,a] = f(x0MLS,tvec(k));
        trajE(1,k) = [x1;a];
    end

    % Get error and cov
    for k = 1:N
        H(2*k-1 : 2*k, :) = jcb(x0MLS,80000,tvec(k));
    end

    P = inv(H'*inv(R)*H);

    figure(); hold on; grid on
    scatter(trajE(1,:),trajE(2,:), 's', 'MarkerFaceColor', [0.5 0 0.8],
'MarkerEdgeColor', [0.5 0 0.8])
    scatter(trajT(1,:),trajT(2,:),100,'m','Color', '#40E0D0')
    legend('MLS Estimate','Ground Truth')
    xlabel('Easting (m)')
    ylabel('Altitude (m)')
    title('MLS Missile Tracking')

    figure(); hold on; grid on
    tiledlayout(2,1)
    nexttile;
    hold on
    title('Error Time History in \xeta')
    exrl = trajE(1,:)-trajT(1,:);
    plot(tvec,exrl,'k',LineWidth=1)
    errorbar(tvec(1),exrl(1),2*sqrt(P(1,1)), -2*sqrt(P(1,1)), 'horizontal', '-o')
    xlabel('Time')
    ylabel('Error')
```

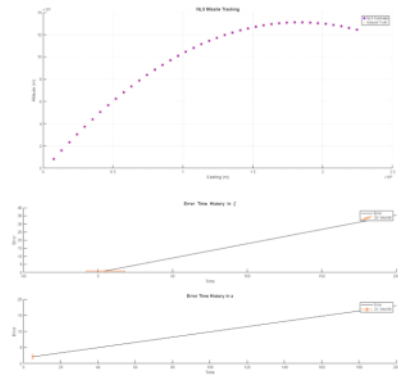
4

```

legend('Error', '2\sigma bounds')
newtile;
hold on
title('Error Time History in a')
err2 = tra2E(2,1)-tra2F(2,1);
plot(tvec,err2,'k',lineWidth=1)
errorbar(tvec(1),err2(1),sqrt(P(2,2)),sqrt(P(2,2)),'-o')
xlabel('Time')
ylabel('Error')
legend('Error', '2\sigma bounds')

```

end



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